

Leaf reflectance and photo- and antioxidant protection in field-grown summer-stressed *Phillyrea angustifolia*. Optical signals of oxidative stress?

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Summary

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- Here we aimed to assess the photo and oxidative stress linked to the Mediterranean summer conditions characterized by drought, heat and high irradiance, and to find possible optical signals of such changes.
- Foliar reflectance was measured in parallel to leaf water content, gas exchange, chlorophyll fluorescence, extent of oxidative stress, and mechanisms of photo- and antioxidative protection in leaves of the evergreen *Phillyrea angustifolia* (*Oleaceae*) plants exposed to Mediterranean field spring-summer conditions.
- Relative leaf water content (RWC) of nonirrigated plants (WS) decreased from 82 to 50% after 48 d without rain, and net photosynthetic rates and stomatal conductances decreased c. 90%, but there was no photo- and oxidative damage. The amounts of the photoprotector zeaxanthin and of the antioxidant α -tocopherol increased up to threefold. The β -carotene concentration decreased to 50%. The reflectance indices WI (water index), PRI (photochemical reflectance index), and BPI (brown pigment index) followed the changes in water content, photosynthetic rates, and oxidant status, respectively.
- These results show that severe drought leads to oxidative stress in field-grown *P. angustifolia* plants, highlights the significance of photo- and antioxidative protection for the survival of drought-stressed plants, and indicates that the changes in oxidant status may be assessed by leaf reflectance (brown pigment index, BPI).

Key words: reflectance, reflectance indices, antioxidants, relative water content (RWC), net photosynthetic rates, summer drought, *Phillyrea angustifolia*.

Abbreviations

AOS, activated oxygen species; α -T, α -tocopherol; β -C, β -carotene; BHT, butylated hydroxytoluene; BPI, brown pigment index; Chl, chlorophyll a + b; DPS, de-epoxidation state of the xanthophyll cycle; $F_v : F_m$, maximum efficiency of photosystem II photochemistry; H, hydration of leaves; HPLC, high-performance liquid chromatography; IR plants, irrigated plants; L, lutein; LMA, leaf mass area; MDA, malondialdehyde; PPFD, photosynthetically active photon flux density; PRI, photochemical reflectance index; PSII, photosystem II; RWC, relative leaf water content; TW, turgid weight; VPD, vapour pressure deficit; WI, water index; WS plants, nonirrigated plants; Z, zeaxanthin.

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Introduction

Soil water availability represents a major environmental constraint under Mediterranean summer conditions (diCastri, 1981). Together with air temperature and solar radiation, it determines the cyclic pattern of vegetation activity. Survival in a Mediterranean climate depends, among other factors, on the ability of plants to make good use of the usually scarce and unpredictable rainfall water during the summer, when water deficit is superimposed on high radiation and high temperatures (Pereira & Chaves, 1993). Many Mediterranean species are evergreen and therefore are ready to use small rain events and short favourable meteorological conditions, but at the same time, they need to further invest in protection to avoid photo- and oxidant damage. Currently, climatic studies of the Mediterranean region have shown a trend of increasing temperature and drought and unpredictability of rainfall in recent decades (Piñol *et al.*, 1998; Houghton *et al.*, 2001; Peñuelas *et al.*, 2002; Peñuelas & Boada, 2003). General circulation models for the next decades forecast even more and stronger droughts for the Mediterranean regions (Houghton *et al.*, 2001).

A number of environmental stresses, including drought, high temperatures and high irradiance, may lead to an imbalance between antioxidant defences and the amount of activated oxygen species (AOS) resulting in oxidative stress (Smirnoff, 1993; Pastori & Foyer, 2002; Xiong *et al.*, 2002). Although AOS are necessary for intra- and intercellular signalling (Doke, 1997; Foyer & Noctor, 1999), accumulation of AOS can cause damage at various levels of organization, including chloroplasts (Halliwell & Gutteridge, 1989; Asada, 1999). Apart from the xanthophyll cycle, photorespiration and other changes in metabolic activity, which may protect the chloroplasts from oxidative damage (Demmig-Adams & Adams, 1996; Kozaki & Takeba, 1996; Eskling *et al.*, 1997; Osmond *et al.*, 1997; Peñuelas & Llusia, 2002), a number of enzymatic and nonenzymatic antioxidants are present in chloroplasts that serve to control oxygen toxicity (Smirnoff, 1993; Foyer *et al.*, 1994; Asada, 1999). Among the latter, tocopherols (vitamin E) and carotenoids play an important role maintaining the integrity of the photosynthetic membranes under oxidative stress (Havaux, 1998; Munné-Bosch & Alegre, 2002a).

Ground level remote sensing allows rapid evaluation of plant and canopy physiological status in a noninvasive way (Peñuelas & Filella, 1998; Gamon & Qiu, 1999). Moreover, remote sensing technology from airborne platforms allows rapid evaluation of vegetation properties in large areas. Many high spectral resolution reflectance vegetation indices have been proposed with the aim of monitoring biomass, phenology and physiological conditions of plants and canopies (Peñuelas & Filella, 1998). Among them, several can be proposed to assess the changes in water status, photosynthetic performance, and oxidative status of the Mediterranean

plants submitted to the stressing drought, high irradiance and high temperatures of the summer. The reflectance water index (WI) has been defined as the ratio $R_{900} : R_{970}$, where R means reflectance and the number stands for the wavelength, and has been shown to be highly correlated with plant water content in several species of trees, shrubs, crops and grasses (Peñuelas *et al.*, 1993, 1994, 1997a). The Photochemical Reflectance Index (PRI, calculated as $[R_{531} - R_{570}] / [R_{570} + R_{531}]$), provides a new approach of estimating photosynthetic performance using spectral reflectance. Absorbance at 531 nm has been found to be related to several processes of dissipation of excess radiation by plants (xanthophyll pigment interconversion, thylakoid ΔpH formation and chloroplast conformational changes) (Bilger & Björkman, 1990; Demmig-Adams, 1990; Demmig & Adams, 1996). The measure of reflectance at this wavelength can thus provide an indirect estimation of photosynthetic performance (Gamon *et al.*, 1990, 1992, 1997; Peñuelas *et al.*, 1994, 1995, 1997b). Several studies have shown a good relationship between PRI and photosynthetic performance when measuring at leaf (Peñuelas *et al.*, 1995; Gamon *et al.*, 1997; Peñuelas *et al.*, 1997b; Richardson & Berlyn, 2002), proximal canopy (Gamon *et al.*, 1992; Filella *et al.*, 1996; Trotter *et al.*, 2002) or at crop or ecosystem scale (Nichol *et al.*, 2000; Peñuelas & Inoue, 2000). Finally, we hypothesised that the oxidative stress linked to the severe summer drought and high temperatures and radiation of the Mediterranean conditions would oxidise compounds such as phenolics to quinones, and would increase the pool of oxidised compounds and pigments often denominated 'brown pigments' (Vaughn & Duke, 1984). The quinonoid products are strongly electrophilic and undergo secondary chemical reactions, which lead to the formation of polymeric brown or black pigments (Nicolas *et al.*, 1994). We define here a new reflectance index based on the reflectance in the far red to assess such brown pigments.

While drought responses in the Mediterranean vegetation have received considerable attention (Chaves *et al.*, 2002), pigment and antioxidant responses, and the optical (reflectance) detection of all these changes have received less attention and are still less understood. Moreover, as far as we are aware, they have not been studied simultaneously. We simultaneously measured water relations, photosynthetic activity, photo- and antioxidative protection, and reflectance of field grown plants of the evergreen species *Phillyrea angustifolia* L. under increasing severe summer drought conditions produced by withholding irrigation. The study was aimed at first characterizing water, photosynthetic, pigment and antioxidant changes in *Phillyrea angustifolia* plants under the progressively drier conditions of the Mediterranean summer, second evaluating the putative relationships among these changes in drought-stressed plants, and third characterizing these physiological responses by plant reflectance changes, especially in the water band at 970 nm, in the green 531 nm band indicative of xanthophyll pigments and photosynthetic performance,

and in the far-red part of the spectrum associated with brown oxidized compounds, which could provide a more specific assessment of oxidative stress. Our general aim was thus to assess the oxidative stress linked to summer drought and to find its possible optical signals.

Materials and Methods

Plant material and treatments

Thirty-six plants of *Phillyrea angustifolia* L. (*Oleaceae*), which were purchased in Garden Bordas (Gavà, Barcelona, Spain), were obtained from seeds and grown in pots of 1–5 l capacity (depending on root biomass as plants grew) containing a mixture of soil:peat:perlite (1 : 1 : 1, by vol.). The pots were maintained in a glasshouse with controlled temperature (24/18°C, day/night) and adequate watering conditions. After growing for 3 yr, plants were transplanted to the experimental fields at the Universitat Autònoma de Barcelona (Bellaterra, Barcelona, NE Spain) on 23 February 2002. Plants were distributed homogeneously in six flat plots of 3 × 3 m² each, so that all plants had the same orientation to the sun. The plots and their surroundings were always maintained clear of vegetation that could have interfered in the growth of *P. angustifolia*. Before the experiment started (18 May 2002), all plants were watered twice a week, so that they received c. 100 mm each month (watering + rainfall). At the beginning of the experiment, plants were 90 ± 5 cm tall ($n = 3$ plants), and their root and leaf d. wt ($n = 3$ plants), respectively.

Two watering conditions were imposed on plants during the experiment (18 May–5 July), and three plots (six plants each) per treatment were used: first irrigated (IR) plants were watered twice a week, so that they received c. 100 mm per month (watering + rainfall), and second plants exposed to water deficit (no irrigation and no rains) (WS plants). Water deficit was imposed by withholding watering to WS plants, and by covering them with a polyvinyl chloride sheet when it rained.

The environmental conditions were monitored with a Davis weather station (Darrera S.A., Esplugues LL., Barcelona, Spain) that was situated at c. 50 m of the experimental plot. Vapour pressure deficit (VPD) was calculated from air temperature (T_a) and relative humidity data according to Nobel (1991).

Plant water status, leaf mass area, the extent of lipid peroxidation (malondialdehyde [MDA] contents), chlorophyll fluorescence, photosynthetic pigments, xanthophyll pigments, α -tocopherol (α -T), and high spectral resolution reflectance in the visible and near-infrared were measured in fully developed young leaves exposed to sun on clear sunny days at midday (at maximum diurnal photosynthetically active photon flux density [PPFD]). For measurements of MDA, photosynthetic pigments, and α -T, leaves were collected, frozen in liquid nitrogen and stored at –20°C until analysis.

Water status and leaf mass area

Leaves were weighted and leaf area was immediately measured using a flatbed scanner (model GT-5000; Epson, Nagano, Japan) and an image-processing program. Then, leaves were re-hydrated for 24 h at 4°C in darkness and subsequently oven-dried for 24 h at 80°C. The relative leaf water content (RWC) was determined as $100 \times (f. wt - d. wt) / (t. wt - d. wt)$, where f. wt is the fresh weight, t. wt is the turgid weight after re-hydrating the leaves, and d. wt is the dry weight after oven-drying the leaves. The leaf mass area (LMA) was determined as d. wt/leaf area. Leaf hydration, H, was calculated as $(f. wt - d. wt) / d. wt$.

Estimation of lipid peroxidation

The extent of lipid peroxidation in leaves was estimated by measuring the amount of MDA by the method described by Hodges *et al.* (1999), which takes into account the possible influence of interfering compounds in the assay for thiobarbituric acid (TBA)-reactive substances. In short, samples were repeatedly extracted with 80 : 20 (v/v) ethanol/water containing 1 ppm butylated hydroxytoluene (BHT) using sonication. After centrifugation, supernatants were pooled and an aliquot of appropriately diluted sample was added to a test tube with an equal volume of either first –TBA solution containing 20% (w/v) trichloroacetic acid and 0.01% (w/v) BHT, or second +TBA solution containing the above plus 0.65% (w/v) TBA. Samples were heated at 95°C for 25 min and, after cooling, absorbance was read at 440 nm, 532 nm, and 600 nm. MDA equivalents (nmol ml^{–1}) were calculated as $10^6 \times ((A - B) / 157\,000)$, where $A = ((Abs\ 532_{+TBA}) - (Abs\ 600_{+TBA}) - (Abs\ 532_{-TBA} - Abs\ 600_{-TBA}))$, and $B = ((Abs\ 440_{+TBA} - Abs\ 600_{+TBA}) \times 0.0571)$.

Chlorophyll fluorescence

Measurements of the maximum efficiency of PSII photochemistry ($F_v : F_m$) were conducted *in situ* on attached leaves with a portable pulse-modulated fluorimeter PAM-2000 (Walz, Effeltrich, Germany). Leaves were dark-adapted with leaf clips for 1 h, which was determined to be sufficient to allow complete relaxation of energy-dependent quenching. The $F_v : F_m$ ratio was calculated as $(F_m - F_o) / F_m$, where F_m and F_o are the maximum and basal fluorescence yields, respectively, of dark-adapted leaves (Genty *et al.*, 1989).

Photosynthetic pigments and α -tocopherol

The extraction and HPLC analyses of photosynthetic pigments and α -T were carried out essentially as described by Munné-Bosch & Alegre (2000). In short, leaves were repeatedly extracted with ice-cold 85 (v/v) and 100% acetone using sonication. Pigments were separated on a Dupont

nonendcapped Zorbax ODS-5 μm column (250×4.6 mm, 20% C, Scharlau, Barcelona, Spain) at 30°C at a flow rate of 1 ml min^{-1} . The solvents consisted of (A) acetonitrile/methanol (85: 15, v/v) and (B) methanol/ethyl acetate (68: 32, v/v). The gradient used was: 0–14 min 100% A, 14–16 min decreasing to 0% A, 16–28 min 0% A, 28–30 min increasing to 100% A, and 30–38 min 100% A. Detection was carried out at 445 nm (Diode array detector 1000S, Applied Biosystems, Foster City, CA). α -T was separated on a Partisil 10 ODS-3 column (250×4.6 mm, Scharlau, Barcelona, Spain) at a flow rate of 1 ml min^{-1} . The solvents consisted of (A) methanol/water (95: 5, v/v) and (B) methanol. The gradient used was: 0–10 min 100% A, 10–15 min decreasing to 0% A, 15–20 min 0% A, 20–23 min increasing to 100% A, and 23–28 min 100% A. α -T was quantified through its absorbance at 283 nm (Diode array detector 1000S, Applied Biosystems). Compounds were identified by their characteristic spectra and by coelution with authentic standards, which were obtained from Fluka (Buchs, Switzerland).

Photosynthetic rates and stomatal conductance

Calibrated LCA-4 IRGA porometers (ADC, Hoddeson, Hertfordshire, UK) were used for determination of CO_2 and H_2O exchange. Intact leaves were clamped in Parkinson Chambers (PLC-2) adapted to ADC-LCA-4. Net carbon dioxide assimilation rates and stomatal conductances were calculated from changes in CO_2 and H_2O exchange according to Von Caemmerer & Farquhar (1981).

Reflectance measurements and indices

Leaf-based reflectance was measured with a UniSpec Spectral Analysis System/Reflectometer (PP Systems, Haverhill, MA, USA), operated with a palmtop PC. Individual leaves from six plants per treatment and date were individually sampled by placing them into a leaf clip (adaxial side upwards) attached to a fibre-optic halogen light source and detector. Four scans per sample were integrated (integration time 10 ms). Reflectance measurements were preceded by a dark scan, and were compared with reflectance from a Spectralon (Labsphere Inc., North Sutton, NH, USA) white standard to obtain percent reflectance. From the reflectance data, various spectral indices and features were derived. The following reflectance indices were computed (R_x means the reflectance at x nm):

- WI (Water Index) = R_{900}/R_{970} (Peñuelas *et al.*, 1993, 1997a; Peñuelas & Filella, 1998)
- PRI (Photochemical reflectance index) = $(R_{531} - R_{570}) / (R_{531} + R_{570})$ (Gamon *et al.*, 1992; Peñuelas *et al.*, 1995)
- BPI (Brown Pigment Index) We defined this index as the difference between the wavelengths where the second derivative of the reflectance in the far-red region intercepts 0, that is the difference between the wavelength with maximum reflectance slope (λ red edge) and the wavelength with the minimum reflectance slope in the near infrared (Fig. 1). This way, the higher brown pigment concentrations will be noticed by the higher BPI values, that is longer wavelength distances from the red edge to the near infrared asymptotic reflectances.

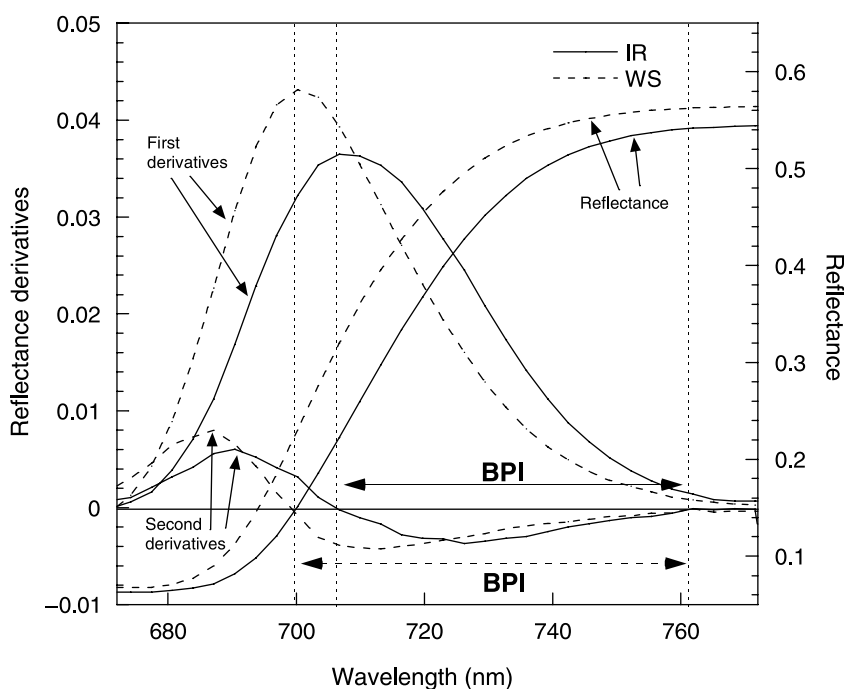


Fig. 1 Reflectance, first derivative reflectance, and second derivative reflectance spectra of an irrigated (IR) and a nonirrigated (WS) plant on 5 July after 48 d without rain. Only the red edge region between 670 and 770 nm is depicted. The brown pigment index (BPI) is calculated as the wavelength difference between the wavelengths where the second derivative of the reflectance in the far-red region intercepts 0, that is the distance between the red edge and the wavelength with the minimum reflectance slope in the near infrared.

Statistical analyses

Statistical differences between measurements on different treatments or on different times were analysed following the Student's *t*-test. Only the average value for each plot was used in the analyses. Differences were considered significant at a probability level of $P < 0.05$. These analyses and additional correlational and regression analyses were conducted using SPSS (Chicago, IL., USA). For regression analyses of mean (average), and therefore of values having known associated variance, a model II regression was used (Coleman *et al.*, 1994). To compute this model II regression, we used the reduced major axis method (Sokal & Rohlf, 1995).

Results and Discussion

Environmental conditions

Environmental conditions during the experimental period (18 May–5 July) were typical of the Mediterranean climate. Maximum diurnal PPFD ranged between 1898 and 2003 $\mu\text{mol m}^{-2} \text{s}^{-1}$ throughout the experiment (Table 1). Maximum diurnal air temperature (T_a) ranged between 26.8 and 33.6°C throughout the study (Table 1). Maximum diurnal VPD ranged from 2.15 to 3.85 KPa (Table 1).

Water content and the reflectance index WI

RWC in IR plants was kept nearly constant around 80% throughout the experiment. Plants exposed to drought in the field experienced progressive leaf water losses. After 48 d of water deficit, RWC in WS plants decreased from 80% to *c.* 50% and the leaf hydration, *H*, from 1.25 to 0.6 g [g d. wt]⁻¹ (Fig. 2).

Reflectance WI distinguished between IR and WS plants, being lower at reduced leaf water content, that is it decreased in WS plants and in day 48 when water content had decreased after several days with no rain and increasing evaporative demand (Fig. 2). WI therefore followed *H* (Figs 2 and 8).

Photosynthetic rates, stomatal conductances and the reflectance index PRI

Net photosynthetic rates decreased 50% in WS plants relative to IR plants the day 15 and *c.* 90% the day 48 when the

atmospheric VPD conditions allowed the IR plants to increase their net photosynthetic rates 2.5 fold those of the day 15 (Fig. 3) Parallel changes were found in stomatal conductances although it must be highlighted that WS plants had almost closed stomata (Fig. 3). Similar trends were found for foliar PRI which this way was therefore a good estimator of net photosynthetic rates (Figs 3 and 8).

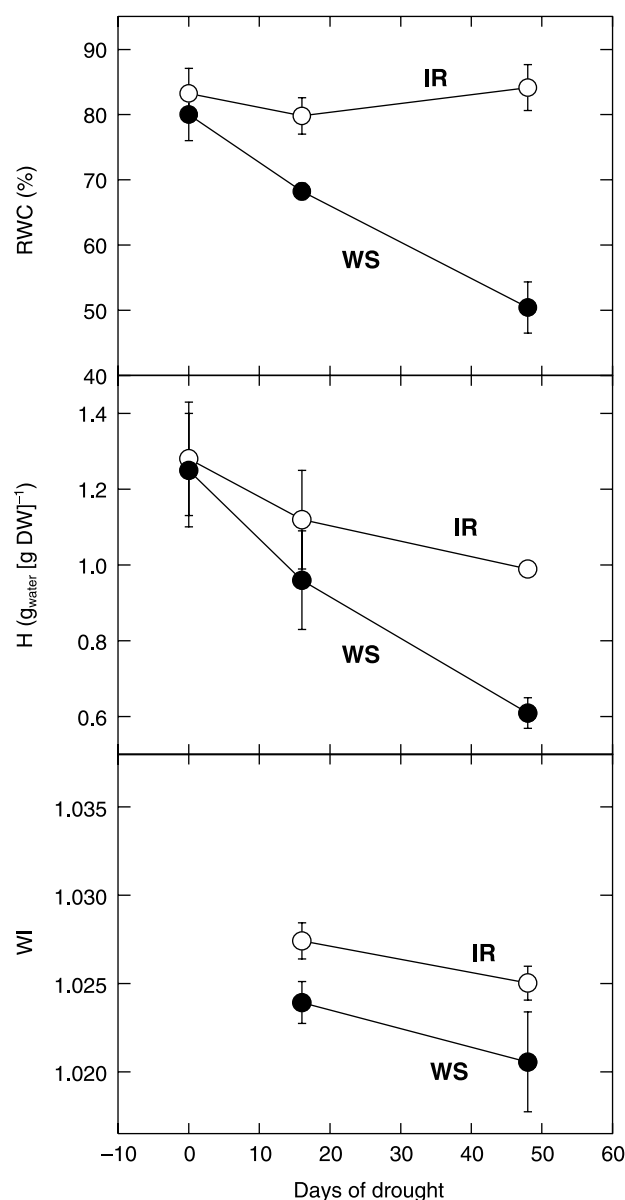


Fig. 2 Relative water content (RWC), foliar hydration (*H*), and reflectance 'Water Index' (WI, R900/R970) of irrigated (IR) and nonirrigated (WS) field-grown *Phillyrea angustifolia* plants during the experiment (from 18 May to 5 July). IR plants were watered twice a week, so that they received *c.* 100 mm of water per month (rainfall + watering). Water deficit was imposed by withholding water to WS plants, and by covering them with a polyvinyl chloride sheet when it rained. Presented data are the means $\pm$ SE ($n = 3$ plot averages of three independent leaves per plot). Leaves were collected at midday (at maximum diurnal photosynthetically active photon flux density [PPFD]).

Table 1 Photosynthetically active photon flux density (PPFD), air temperature (T_{air}), and vapour pressure deficit (VPD) at midday during the measurement days between May and July 2002

	PPFD ($\mu\text{mol m}^{-2} \text{s}^{-1}$)	T_{air} ($^{\circ}\text{C}$)	VPD (KPa)
18 May	1898	26.8	2.20
3 June	1922	33.6	3.85
5 July	2003	30.6	2.15

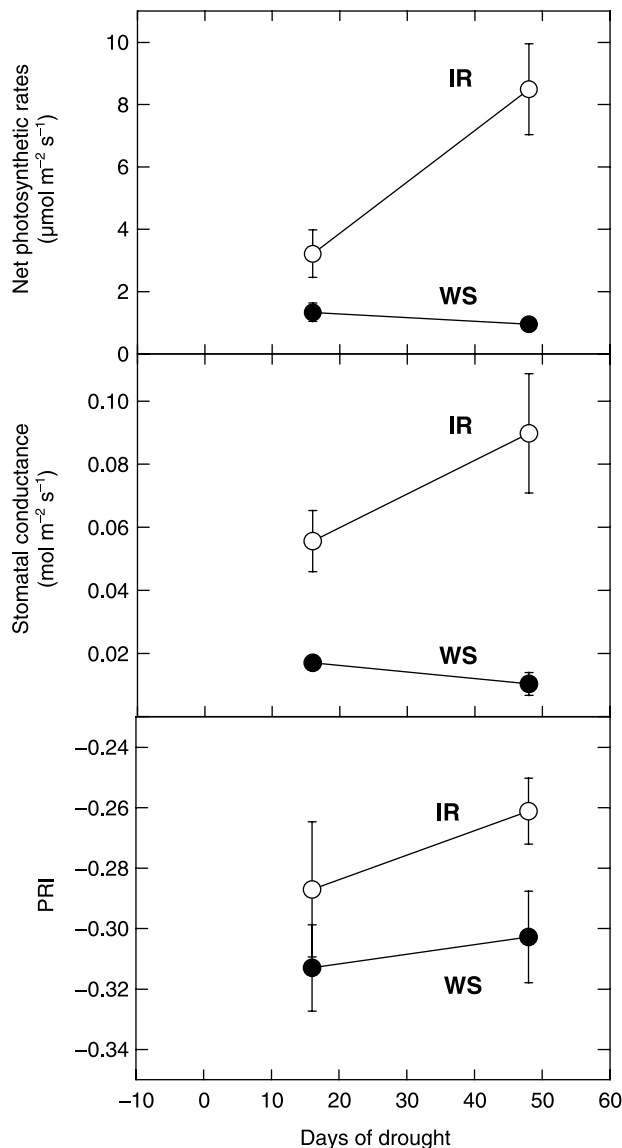


Fig. 3 Net photosynthetic rates, stomatal conductance and photochemical reflectance index (PRI, $R531 - R570/R531 + R570$) in leaves of irrigated (IR) and nonirrigated (WS) *Phillyrea angustifolia* plants. IR plants were watered twice a week, so that they received c. 100 mm of water per month (rainfall + watering). Water deficit was imposed by withholding water to WS plants, and by covering them with a polyvinyl chloride sheet when it rained. Data correspond to the means $\pm$ SE ($n = 3$ plot averages of two independent leaves per plot). Measurements were made on leaves collected at midday.

Photo- and antioxidant protection and the reflectance index BPI

Although after the 48-d period of study, Chl levels decreased by c. 20% in IR plants and 25% in WS plants (Fig. 4), they did not present significant differences between IR and WS plants. No significant differences were found either for their leaf mass per area (data not shown), their maximum efficiency of photosystem II photochemistry ($F_v : F_m$ ratio) or their

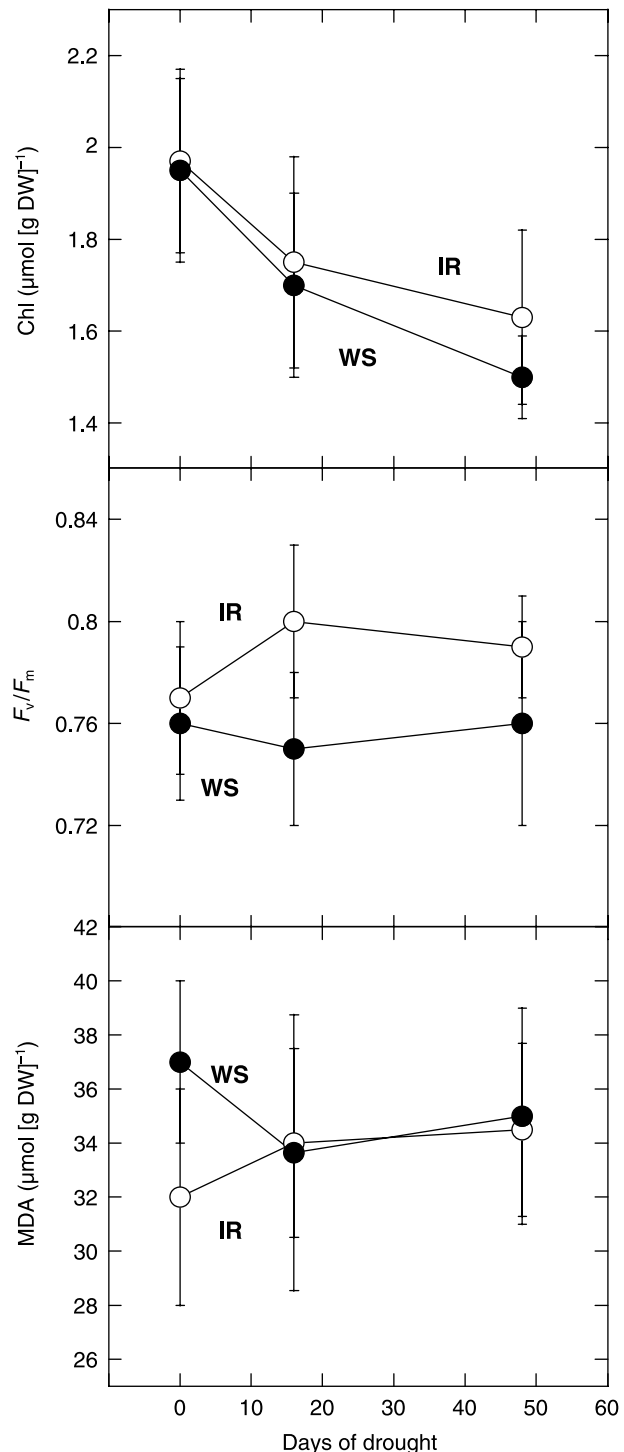


Fig. 4 Chlorophyll a + b (Chl) levels, maximum efficiency of photosystem II photochemistry ($F_v : F_m$), and malondialdehyde (MDA) accumulation in leaves of irrigated (IR) and nonirrigated (WS) *Phillyrea angustifolia* plants. IR plants were watered twice a week, so that they received c. 100 mm of water per month (rainfall + watering). Water deficit was imposed by withholding water to WS plants, and by covering them with a polyvinyl chloride sheet when it rained. Data correspond to the means $\pm$ SE ($n = 3$ plot averages of two independent leaves per plot). Measurements were made on leaves collected at midday.

malondialdehyde concentrations, which did not change significantly either throughout the period of study or between treatment levels (Fig. 4). These results indicate an absence of a significant photoinhibitory or oxidative damage to the photosynthetic apparatus, and therefore that the photo- and antioxidative protection mechanisms were able to protect the plants under these typical summer drought conditions.

Mechanisms of photo- and antioxidant protection in chloroplasts (xanthophyll cycle and levels of lipophilic antioxidants) during drought were evaluated. The amounts of zeaxanthin of WS plants increased significantly up to 2.5 fold at RWCs of *c.* 50% at day 48, which caused up to *c.* 75% increase in the total amount of xanthophylls pigments and in the de-epoxidation state of the xanthophyll cycle (DPS) (Fig. 5). The total amount of the xanthophyll cycle components (VZA) remained unaltered in IR plants, and increased by *c.* 0.06 $\mu\text{mol} [\text{g d. wt}]^{-1}$ in WS plants (Fig. 5). This was mainly due to the large increases in zeaxanthin, which increased in these plants by *c.* 0.075 $\mu\text{mol} [\text{g d. wt}]^{-1}$. Enhanced zeaxanthin formation was therefore not only due to the de-epoxidation of the violaxanthin present in irrigated (IR) plants, but also to the *de novo* synthesis of carotenoids under stress. Zeaxanthin attained maximum levels in WS plants, in which zeaxanthin accounted for 65% of the xanthophyll cycle components present in leaves. The de-epoxidation state of the xanthophyll cycle was *c.* 0.45 in IR plants and reached its maximum levels (*c.* 0.75) in WS plants on day 48 (Fig. 5).

Chloroplastic lipophilic antioxidants (carotenoids and tocopherols) differed in their response to drought and recovery in *P. angustifolia* (Fig. 6). While lutein levels did not show significant alterations throughout the experiment in IR and WS plants, β -carotene (β -C) decreased and α -tocopherol (α -T) increased, as drought progressed. β -C decreased with time 20% in IR plants and 45% in WS plants (Fig. 6), and α -T increased up to 2–3-fold, at RWCs of *c.* 50% of the WS plants on day 48, whereas it remained constant in the IR plants (Fig. 6).

The reflectance index BPI, indicative of brown pigments (oxidized compounds), increased *c.* 8–10% in the WS plants on day 48 (Fig. 6) and therefore was negatively correlated with the β -carotene concentrations (Fig. 7). This index also presented similar patterns to Z, DPS, and α -T, that is to photo- and antioxidant protection.

Water and oxidative status under increasing summer drought

General ecophysiological responses to increasing summer drought in Mediterranean plants have been partially elucidated in numerous studies that have shown similar leaf responses to those described here of decreasing leaf water content, chlorophyll concentrations, net photosynthetic rates and stomatal conductances with increasing drought (Tenhunen *et al.*, 1990; Peñuelas *et al.*, 1998). However, we show here the importance of coordinated physiological responses including photo- and antioxidant protective mechanisms.

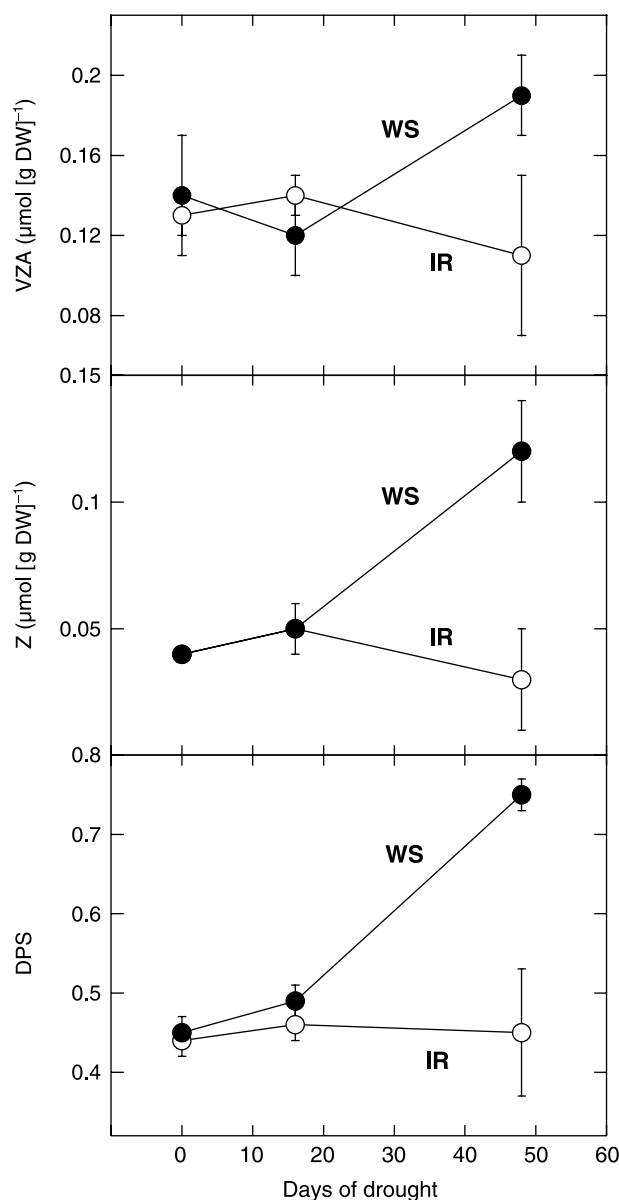


Fig. 5 Xanthophyll cycle pool (violaxanthin + zeaxanthin + antheraxanthin [VZA]), zeaxanthin (Z) levels, and de-epoxidation state of the xanthophyll cycle (DPS) in leaves of irrigated (IR) and nonirrigated (WS) *Phillyrea angustifolia* plants. IR plants were watered twice a week, so that they received *c.* 100 mm of water per month (rainfall + watering). Water deficit was imposed by withholding water to WS plants, and by covering them with a polyvinyl chloride sheet when it rained. DPS was calculated as $(Z + 0.5 A)/(VZA)$. Data correspond to the means $\pm$ SE ($n = 3$ plot averages of two independent leaves per plot). Measurements were made on leaves collected at midday.

Enhanced de-epoxidation of the xanthophyll cycle and antioxidant levels have been associated with increased drought stress resistance in several Mediterranean species (Chaves *et al.*, 2002). The mechanisms that *Phillyrea angustifolia* plants have evolved to withstand stress are of special interest because

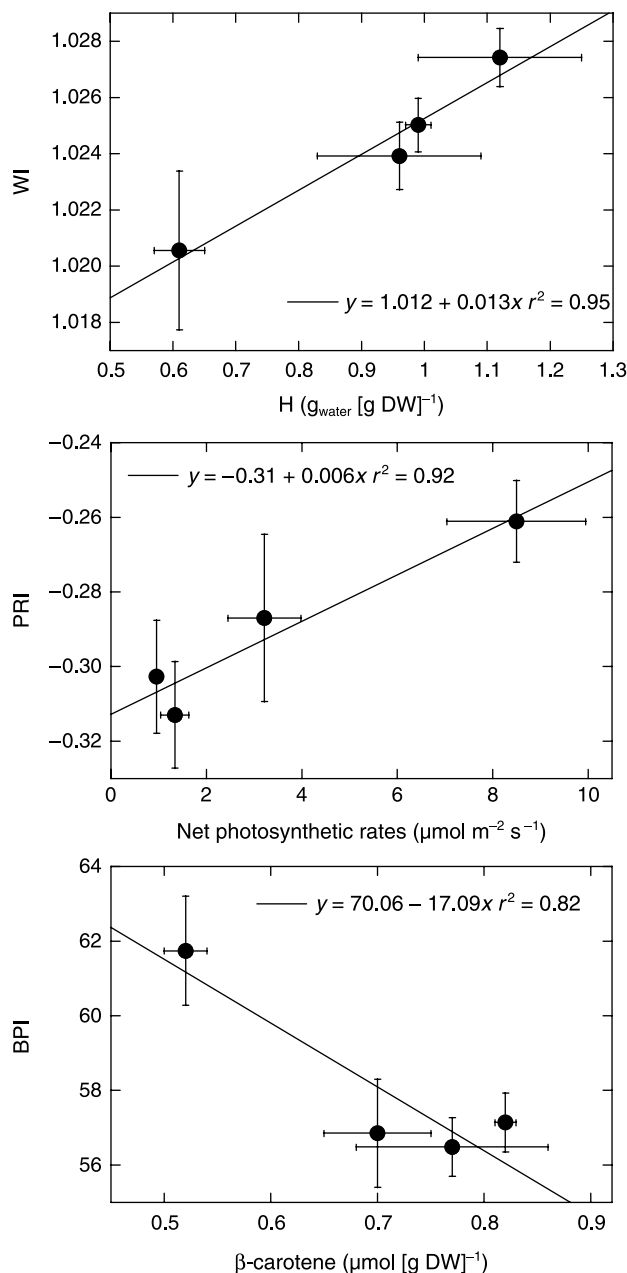
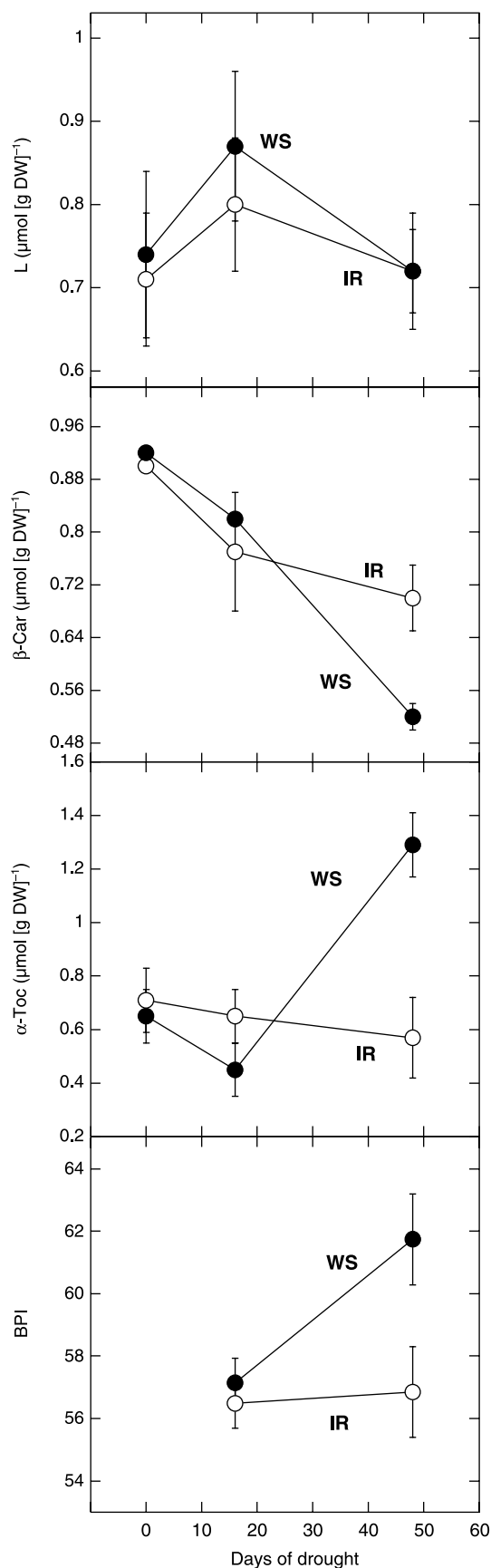


Fig. 7 Relationships of the leaf hydration (H), the net photosynthetic rates, and the β-carotene concentrations with the corresponding reflectance indices WI, PRI and BPI, respectively. Data correspond to the means $\pm$ SE ($n = 3$ plot averages of two independent leaves per plot). Measurements were made on leaves collected at midday.

Fig. 6 α-tocopherol (α-T), Lutein (L), β-carotene (β-C) and Brown Pigment Index (BPI) levels in leaves of irrigated (IR) and nonirrigated (WS) *Phillyrea angustifolia* plants. IR plants were watered twice a week, so that they received c. 100 mm of water per month (rainfall + watering). Water deficit was imposed by withholding water to WS plants, and by covering them with a polyvinyl chloride sheet when it rained. (BPI was defined as the wavelength distance between the red edge and the near infrared reflectance stabilization). Data correspond to the means $\pm$ SE ($n = 3$ plot averages of two independent leaves per plot). Measurements were made on leaves collected at midday.

these shrubs are typical of the Mediterranean climate, and can withstand a severe summer. In the present study, dissipation of excess excitation energy in the xanthophyll cycle and increased α -T levels (which detoxify AOS and lipid peroxyl radicals) seem to afford protection to the photosynthetic apparatus in drought-stressed *P. angustifolia* plants. Thus, this species did not suffer from photoinhibitory damage to its photosynthetic apparatus (as indicated by the absence of significant changes in $F_v : F_m$ and MDA levels), at least at the drought stress levels here tested (RWCs above 50%). The slight trend of Chl breakdown with the increasing summer drought (Fig. 4), decreases leaf light absorption, and thus decreases Chl molecules as a source for potentially harmful singlet oxygen production (Halliwell & Gutteridge, 1989) and increases the photo- and antioxidative protection per amounts of photons absorbed (Kyparissis *et al.*, 1995; Havaux *et al.*, 1998; Munné-Bosch & Alegre, 2000). As a result, plants did not suffer from photo or oxidative damage as indicated by the absence of any change or difference in the $F_v : F_m$ ratio and MDA foliar concentrations (Fig. 4). These results therefore indicate that *P. angustifolia* leaves can avoid damage to the photosynthetic apparatus by changing pigment and antioxidant levels in response to drought-induced oxidative stress.

Optical signals

These ecophysiological changes were followed by reflectance changes, being WI a good index for water content (Peñuelas *et al.*, 1993, 1997a), and PRI a good index for photosynthetic performance (Gamon *et al.*, 1992; Peñuelas *et al.*, 1994, 1995, 1997b), confirming previous studies (Peñuelas & Filella, 1998). However, what is novel in these results is that they show the possibility that the brown pigment index, BPI, here defined, might become good optical signals for oxidative stress since it was positively correlated with Z and α -T concentrations, that is with increased photo- and antioxidant protection, and with β -carotene degradation. This is what should be expected from an index defined to assess brown pigments, that is polymers of oxidized compounds such as quinones. These quinones are produced as a result of the oxidation of phenolic compounds in a reaction catalyzed by the polyphenol oxidase (PPO). The exact physiological role of PPO in plant cells is not well established, but PPO has been implicated to function in the oxygen scavenging in chloroplast (Trebst & Depka, 1995). The *o*-quinones are highly reactive in plant cells and are subsequently involved in the formation of polymeric dark-colored pigment deposits (Nicolas *et al.*, 1994). In any case, the concentrations of brown pigments will have to be measured in future studies to test these relationships.

Conclusions

In conclusion, we have demonstrated first that the evergreen *P. angustifolia* plants activate several mechanisms of photo- and

antioxidant protection to withstand drought stress during the Mediterranean summer, second that, when the drought is not extremely severe, these mechanisms are able to protect the tissues from lipid peroxidation, and to allow the maintenance of high photosynthetic rates if meteorological conditions are adequate, and third that all these processes can be reasonably followed by the optical sensing provided by reflectance indices in the visible, the far red and the near infrared. However, further studies on these issues are warranted before these optical signals, especially the BPI here defined, can be used in a broad range of species and conditions.

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References

- Asada K. 1999. The water-water cycle in chloroplasts: scavenging of active oxygens and dissipation of excess photons. *Annual Review of Plant Physiology and Plant Molecular Biology* 50: 601–639.
- Bilger W, Björkman O. 1990. Role of the xanthophyll cycle in photoprotection elucidated by measurements of light-induced absorbance changes, fluorescence and photosynthesis in leaves of *Hedera canariensis* Willd. *Photosynthesis Research* 25: 173–185.
- diCasteri F. 1981. Mediterranean-type shrublands of the world. In: diCasteri F, Goodall DW, Specht RL, eds. *Mediterranean-type shrublands*. Amsterdam, The Netherlands: Elsevier Scientific Publishing, 1–52.
- Chaves MM, Pereira JS, Maroco J, Rodrigues ML, Ricardo CPP, Osorio ML, Carvalho I, Faria T, Pinheiro C. 2002. How plants cope with water stress in the field? Photosynthesis and growth. *Annals of Botany* 89: 907–916.
- Coleman JS, McConaughay KDM, Ackerly DD. 1994. Interpreting phenotypic variation in plants. *Trends in Ecology and Evolution* 9: 187–191.
- Demmig-Adams B. 1990. Carotenoids and photoprotection: a role for the xanthophyll zeaxanthin. *Biochimica Biophysica Acta* 1020: 1–24.
- Demmig-Adams B, Adams WW III. 1996. The role of xanthophyll cycle carotenoids in the protection of photosynthesis. *Trends in Plant Science* 1: 21–26.
- Doke N. 1997. The oxidative burst: role in signal transduction and plant stress. In: Scandalios SG, ed. *Oxidative stress and the molecular biology of antioxidant defences*. New York, USA: Cold Spring Harbor Laboratory Press, 785–813.
- Eskling M, Arvidsson PO, Akerlund HE. 1997. The xanthophyll cycle, its regulation and components. *Physiologia Plantarum* 100: 806–816.
- Filella I, Amaro T, Araus JL, Peñuelas J. 1996. Relationship between photosynthetic radiation-use efficiency of barley canopies and the photochemical reflectance index (PRI). *Physiologia Plantarum* 96: 211–216.
- Foyer CH, Lelandais M, Kunert KJ. 1994. Photooxidative stress in plants. *Physiologia Plantarum* 92: 696–717.
- Foyer CH, Noctor G. 1999. Leaves in the dark see the light. *Science* 284: 599–601.
- Gamon J, Field C, Björkman O, Artree A, Peñuelas J. 1990. Remote sensing of the xanthophyll cycle and chlorophyll fluorescence in sunflower leaves and canopies. *Oecologia* 85: 1–7.

- Gamon JA, Peñuelas J, Field CB. 1992. A narrow-waveband spectral index that tracks diurnal changes in photosynthetic efficiency. *Remote Sensing of Environment* 41: 35–44.
- Gamon JA, Qiu H. 1999. Ecological applications of remote sensing at multiple scales. In: Pugnaire FI, Valladares F, eds. *Handbook of functional plant ecology*. New York, USA: Marcel Dekker, 805–846.
- Gamon JA, Serrano L, Surfus JS. 1997. The photochemical reflectance index: an optical indicator of photosynthesis radiation use efficiency across species, functional types and nutrient levels. *Oecologia* 112: 492–501.
- Genty B, Briantais JM, Baker NR. 1989. The relationship between the quantum yield of photosynthetic electron transport and quenching of chlorophyll fluorescence. *Biochimica Biophysica Acta* 990: 87–92.
- Halliwell B, Gutteridge JMC. 1989. *Free radicals in biology and medicine*. Oxford, UK: Oxford University Press.
- Havaux M. 1998. Carotenoids as membrane stabilizers in chloroplasts. *Trends in Plant Science* 3: 147–151.
- Havaux M, Tardy F, Lemoine Y. 1998. Photosynthetic light-harvesting function of carotenoids in higher-plant leaves exposed to high light irradiances. *Planta* 205: 242–250.
- Hodges DM, DeLong JM, Forney CF, Prange RK. 1999. Improving the thiobarbituric acid-reactive-substances assay for estimating lipid peroxidation in plant tissues containing anthocyanin and other interfering compounds. *Planta* 207: 604–611.
- Houghton JT, Ding Y, Griggs DJ, Noguer M, Van der Linden PJ, Dai X, Maskell K, Johnson CA. 2001. *IPPC Climate change 2001: The scientific basis. Contribution of Working Group I Third assessment report of the intergovernmental panel of climate change*. Cambridge, UK: Cambridge University Press.
- Kozaki A, Takeba G. 1996. Photorespiration protects C₃ plants from photooxidation. *Nature* 384: 557–560.
- Kyparissis A, Petropoulou Y, Manetas Y. 1995. Summer survival of leaves in a soft-leaved shrub (*Phlomis fruticosa* L., Labiatae) under Mediterranean field conditions: avoidance of photoinhibitory damage through decreased chlorophyll contents. *Journal of Experimental Botany* 46: 1825–1831.
- Munné-Bosch S, Alegre L. 2000. Changes in carotenoids, tocopherols and diterpenes during drought and recovery, and the biological significance of chlorophyll loss in *Rosmarinus officinalis* plants. *Planta* 210: 925–931.
- Munné-Bosch S, Alegre L. 2002a. The function of tocopherols and tocotrienols in plants. *Critical Reviews in Plant Sciences* 21: 31–57.
- Nichol CJ, Huemmrich KF, Black TA, Jarvis PG, Walthall CL, Grace J, Hall FG. 2000. Remote sensing of photosynthetic-light-use efficiency of boreal forest. *Agricultural and Forest Meteorology* 101: 131–142.
- Nicolas J, Richard-Forget FC, Goupy PM, Amiot MJ, Aubert SY. 1994. Enzymatic browning reactions in apple and apple products. *Critical Reviews in Food Sciences and Nutrition* 34: 109–157.
- Nobel PS. 1991. *Physicochemical and environmental plant physiology*. San Diego, CA, USA: Academic Press.
- Osmond B, Badger M, Maxwell K, Björkman O, Leegod R. 1997. Too many photons: photorespiration, photoinhibition and photooxidation. *Trends in Plant Science* 2: 119–120.
- Pastori G, Foyer CH. 2002. Common components, networks, and pathways of cross-tolerance to stress. The central role of 'redox' and abscisic acid-mediated controls. *Plant Physiology* 129: 460–468.
- Peñuelas J, Boada M. 2003. A global change-induced biome shift in the Montseny mountains (NE Spain). *Global Change Biology* 9: 131–140.
- Peñuelas J, Filella I. 1998. Visible and near-infrared reflectance techniques for diagnosing plant physiological status. *Trends in Plant Science* 3: 151–156.
- Peñuelas J, Filella I, Comas P. 2002. Changed plant and animal life cycles from 1952 to 2000 in the Mediterranean region. *Global Change Biology* 8: 531–544.
- Peñuelas J, Filella I, Gamon JA. 1995. Assessment of photosynthetic radiation-use efficiency with spectral reflectance. *New Phytologist* 131: 291–296.
- Peñuelas J, Filella I, Llusia J, Siscart D, Piñol J. 1998. Comparative field study of spring and summer leaf gas exchange and photobiology of the mediterranean trees *Quercus ilex* and *Phillyrea latifolia*. *Journal of Experimental Botany* 49: 229–238.
- Peñuelas J, Filella I, Serrano L, Biel C, Save R. 1993. The reflectance at the 950–970 nm region as an indicator of plant water status. *International Journal of Remote Sensing* 14: 1887–1905.
- Peñuelas J, Gamon J, Fredeen A, Merino J, Field C. 1994. Reflectance indices associated with physiological changes in nitrogen- and water-limited sunflower leaves. *Remote Sensing of Environment* 48: 135–146.
- Peñuelas J, Inoue Y. 2000. Reflectance assessment of canopy CO₂ uptake. *International Journal of Remote Sensing* 21: 3353–3356.
- Peñuelas J, Llusia J. 2002. Linking photorespiration, monoterpenes and thermotolerance in *Quercus*. *New Phytologist* 155: 227–237.
- Peñuelas J, Llusia J, Piñol J, Filella I. 1997b. Photochemical reflectance index and leaf photosynthetic radiation-use-efficiency assessment in Mediterranean trees. *International Journal of Remote Sensing* 13: 2863–2868.
- Peñuelas J, Piñol J, Ogaya R, Filella I. 1997a. Estimation of plant water content by the reflectance Water Index WI (R900/R970). *International Journal of Remote Sensing* 18: 2869–2875.
- Pereira JS, Chaves MM. 1993. Plant water deficits in Mediterranean ecosystems. In: Smith JAC, Griffiths H, eds. *Water deficits plant responses from cell to community*. Oxford, UK: Bios Scientific Publishers, 237–251.
- Piñol J, Terrades J, Lloret F. 1998. Climate warming, wildfire hazard, and wildfire occurrence in coastal eastern Spain. *Climatic Change* 38: 345–357.
- Richardson AD, Berlyn GP. 2002. Spectral reflectance and photosynthetic properties of *Betula papyrifera* (Betulaceae) leaves along an elevational gradient on Mt. Mansfield, Vermont, USA. *American Journal of Botany* 89: 88–94.
- Smirnoff N. 1993. The role of active oxygen in the response of plants to water deficit and desiccation. *New Phytologist* 125: 27–58.
- Sokal RR, Rohlf FJ. 1995. *Biometry the principles and practice of statistics in biological research*, 3rd edn. New York, USA: W. H. Freeman.
- Tenhunen JD, Sala Serra A, Harley PC, Dougherty RL, Reynolds JF. 1990. Factors influencing carbon fixation and water use by mediterranean sclerophyll shrubs during summer drought. *Oecologia* 82: 381–393.
- Trebst A, Depka B. 1995. Polyphenol oxidase and photosynthesis research. *Photosynthesis Research* 46: 41–44.
- Trotter GM, Whitehead D, Pinkney EJ. 2002. The photochemical reflectance index as a measure of photosynthetic light use efficiency for plants with varying foliar nitrogen contents. *International Journal of Remote Sensing* 6: 1207–1212.
- Vaughn KC, Duke SO. 1984. Function of polyphenol oxidase in higher plants. *Physiologia Plantarum* 60: 106–112.
- Von Caemmerer S, Farquhar GD. 1981. Some relations between the biochemistry of photosynthesis and the gas exchange of leaves. *Planta* 153: 376–387.
- Xiong L, Schumaker KS, Zhu J. 2002. Cell signaling during cold, drought, and salt stress. *Plant Cell* 14: S165–S183.